

Use-Case Flow Specification

Use Case: Customize Monthly Box

Use Case ID	UC-002
Use Case Name	Customize Monthly Box
Primary Actor	Subscriber
Secondary Actor(s)	System (Recommendation Engine, Fulfillment Service)
Related Requirements	FR-001, FR-002, FR-005
Description	Allows a subscriber to swap, add, or remove items in their upcoming monthly subscription box based on personal preferences, provided the customization window is still open (≥ 48 hours before billing renewal).

Preconditions

1. The subscriber is registered and has an active (non-cancelled) subscription.
2. The subscriber is authenticated and logged into the portal.
3. The next billing cycle is scheduled, and the current date/time is **at least 48 hours before** the monthly billing renewal date.
4. A default box has been generated for the upcoming cycle based on the subscriber's preference tags.

Postconditions

1. The subscriber's monthly box contents are updated to reflect the new selections.
2. The fulfillment manifest is refreshed to include the updated item list for this subscriber.
3. A confirmation notification (email + in-app) is sent to the subscriber summarizing the changes.
4. The inventory count for swapped-out items is restored and swapped-in items are reserved.

Main Success Scenario

Step	Actor	Action
1	Subscriber	Navigates to the "My Upcoming Box" page from the dashboard.
2	System	«include» Validate Customization Window: Checks that the current time is ≥ 48 hours before the billing renewal date. Validation passes.

Step	Actor	Action
3	System	«include» Apply Preference-Based Recommendations: Loads the current box contents alongside personalized product recommendations based on the subscriber's preference tags.
4	System	Displays the current box items, available alternative products (with preference-match scores), and the customization deadline.
5	Subscriber	Reviews the current box and selects one or more items to swap. For each swap, the subscriber picks a replacement from the recommended alternatives or browses the full catalog.
6	System	For each swap request, validates that the replacement item is in stock and compatible with the subscriber's subscription tier.
7	System	Updates the box preview in real time showing the new item(s), adjusted box summary, and any price difference (if applicable).
8	Subscriber	Reviews the updated box preview and confirms the changes by clicking "Save My Box."
9	System	Persists the updated box selections to the database. Releases inventory hold on removed items and reserves inventory for the newly selected items.
10	System	Refreshes the fulfillment manifest for the current billing cycle to reflect the subscriber's updated box contents.
11	System	Sends a confirmation notification (email + in-app) to the subscriber with the finalized box summary and expected delivery date.
12	System	Displays a success message: "Your box has been updated successfully!"

Alternate Flow: Customization Window Closed

Step	Actor	Action
2a	System	At Step 2, the system determines that the current time is less than 48 hours before the billing renewal date. The customization window has closed.
2b	System	Displays a message: "The customization window for this cycle has closed. Your current box will be shipped as-is. You can still customize your next month's box after the current cycle renews."
2c	System	Shows the current (locked) box contents in read-only mode, along with the date when customization for the next cycle will open.
2d	Subscriber	Acknowledges the message. May choose to navigate to subscription settings to skip the current cycle (if skip window is also still open), or simply exit.

Exceptions

ID	Condition	System Response
E1	Selected replacement item is out of stock.	System displays "This item is currently unavailable" and suggests the next-best alternative from the recommendation engine.
E2	Database or network error during save.	System displays "We couldn't save your changes. Please try again." Changes are rolled back; inventory holds are not modified.
E3	Session expires during customization.	System redirects to the login page. Unsaved changes are stored temporarily and restored upon re-authentication.

Business Rules

- **BR-01:** A subscriber may customize their box an unlimited number of times within the customization window. Only the final saved state is sent to fulfillment.
- **BR-02:** The customization window opens immediately after the previous cycle's billing renewal and closes 48 hours before the next renewal.
- **BR-03:** Price differences from item swaps (e.g., premium items) are reflected as add-on charges or credits on the next billing statement.